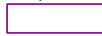
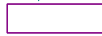


Daisy



File: Seed3-DevKit-Desktop_1.kicad_sch

Peripherals



File: Seed3-DevKit-Desktop_2.kicad_sch

Audio



File: audio.kicad_sch

POWER

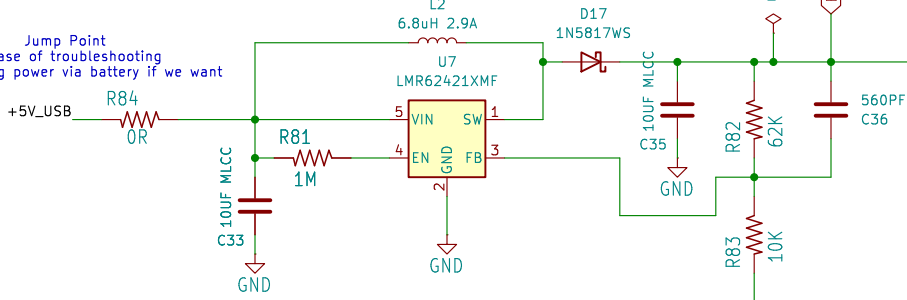
Selecting R2 with formula:
 $R2 = (V_{out}/V_{ref} - 1) \times R1$
 Where $V_{ref} = 1.255V$

For V_{out} : where $R_1 = 10K$ and $R_2 = 62K$
 $(R_2/R_1)+1 * V_{ref} = V_{out}$

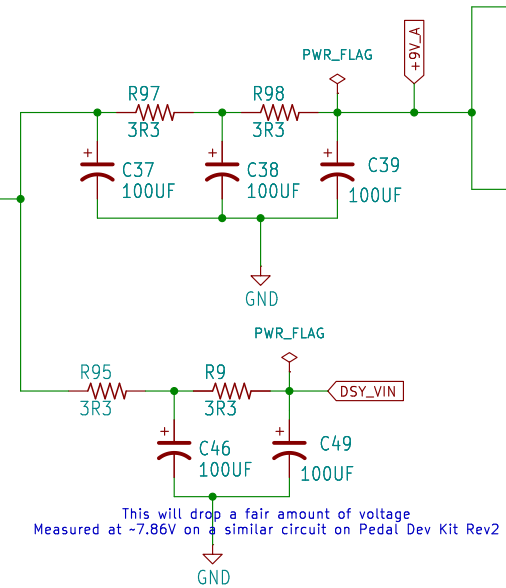
$$V_{out} = 9.036V$$

(or if my algebra was wrong, it would be.. 8.781V)
Either is totally fine .. and they're pretty close in this case.

Jump Point
In case of troubleshooting
or for testing power via battery if we want



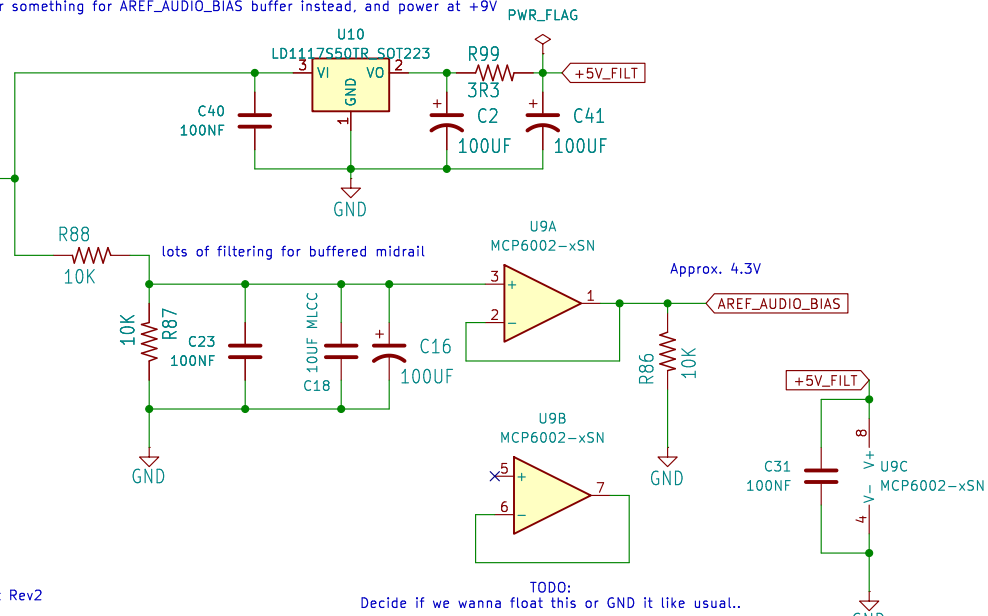
This will drop a fair amount of voltage
Measured at ~8.6V on similar circuit on Pedal Dev Kit Rev2



- At the moment, the +5V is powering
 - Headphone Amp
 - DAC output op-amp
 - Reference voltage for CV inputs
 - 6002 buffer for AREF_AUDIO_BIAS

We can remove this, and simplify slightly if we:

- Change DAC output op-amp to a rail-to-rail opamp that supports higher voltage (i.e. OPA1652)
- Power DAC output op-amp from +9V_A
- Add LM4040 for +5V reference for CV inputs
- Use TL074 or something for AREF_AUDIO_BIAS buffer instead, and power at +9V



Decide if we wanna float this or GND it like usual..

open hardware

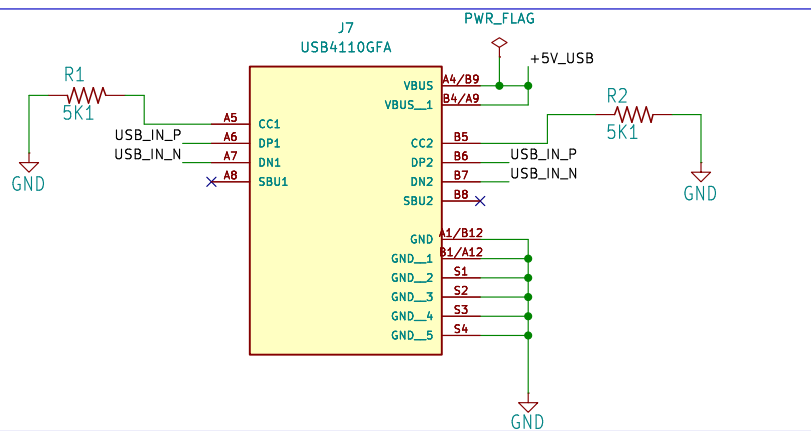
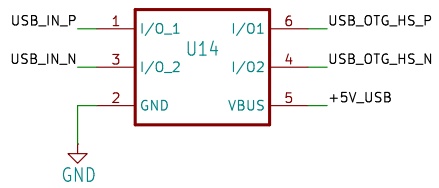
QR1



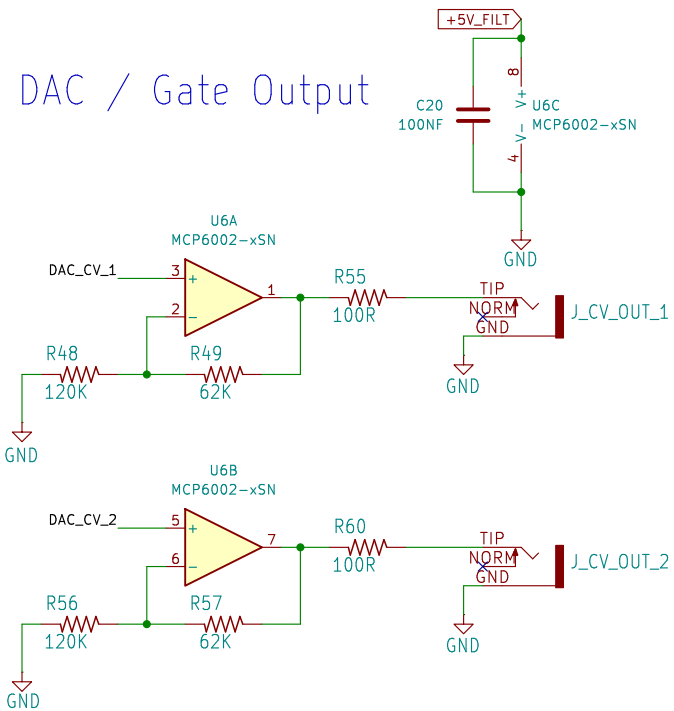
○ ○ ○

- H1 MOUNT-PAD-ROUND3.5
- H2 MOUNT-PAD-ROUND3.5
- H3 MOUNT-PAD-ROUND3.5
- H4 MOUNT-PAD-ROUND3.5

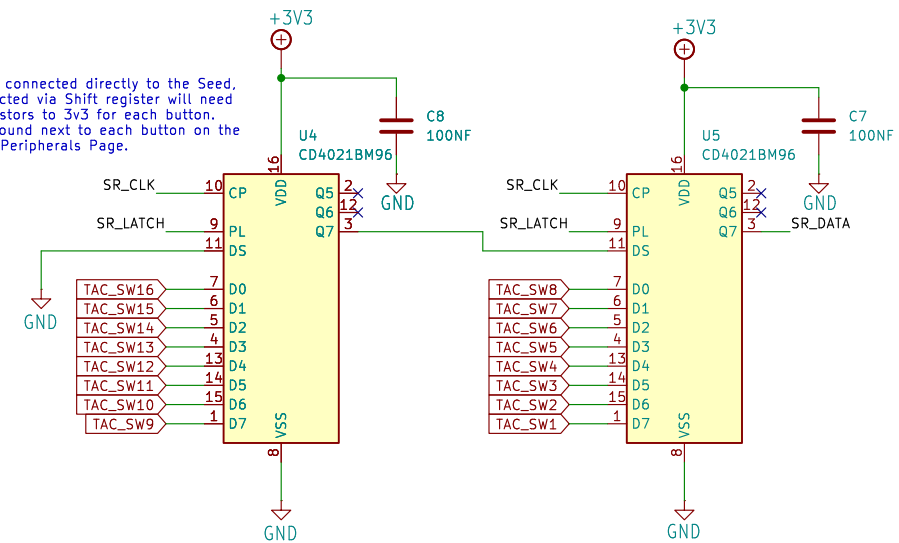
USB C



DAC / Gate Output

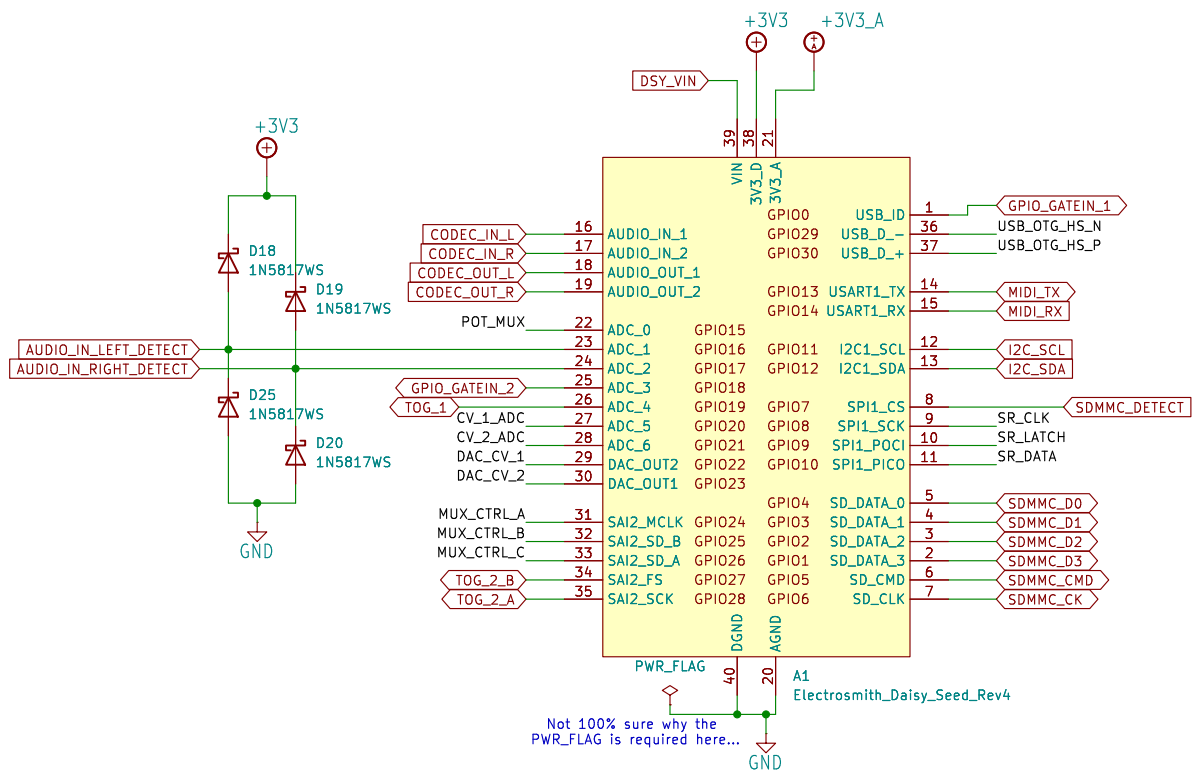
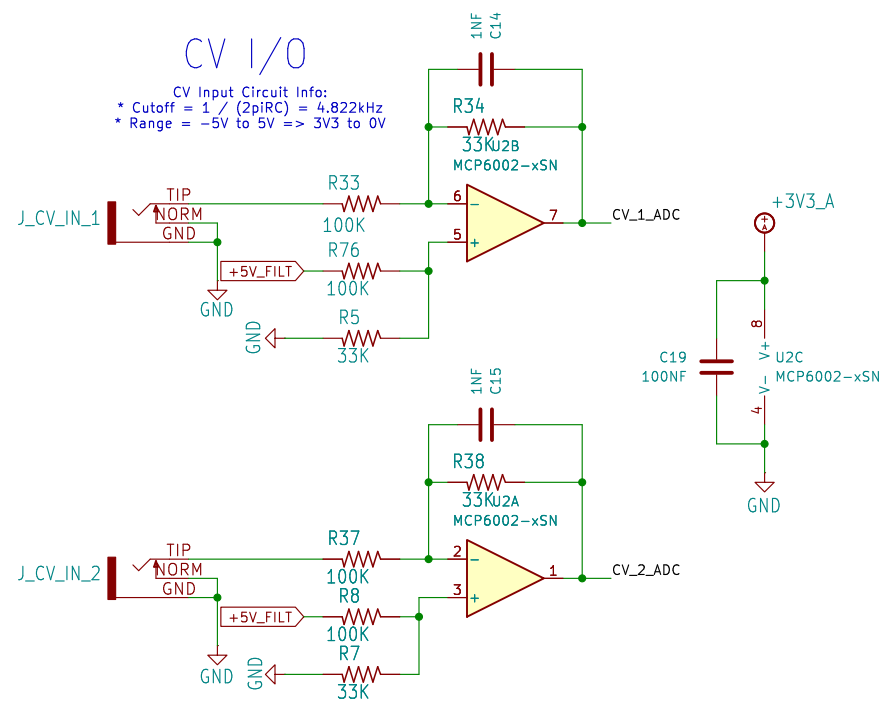


Unlike buttons connected directly to the Seed, buttons connected via Shift register will need pull up resistors to 3v3 for each button. These can be found next to each button on the Peripherals Page.



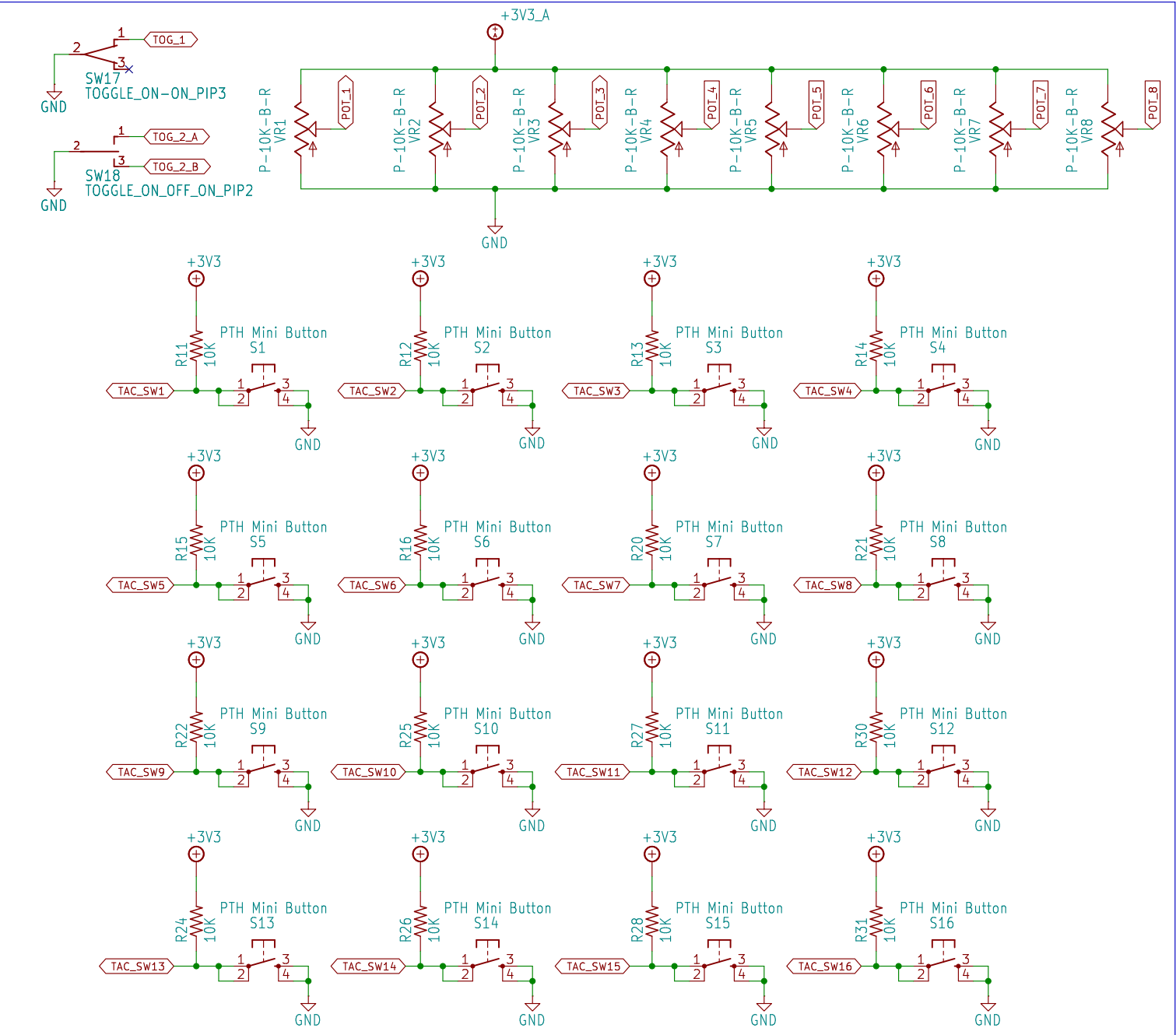
CV 1/0

CV Input Circuit Info:
 * Cutoff = $1 / (2\pi RC) = 4.822\text{kHz}$
 * Range = -5V to $5\text{V} \Rightarrow 3\text{V3}$ to 0V

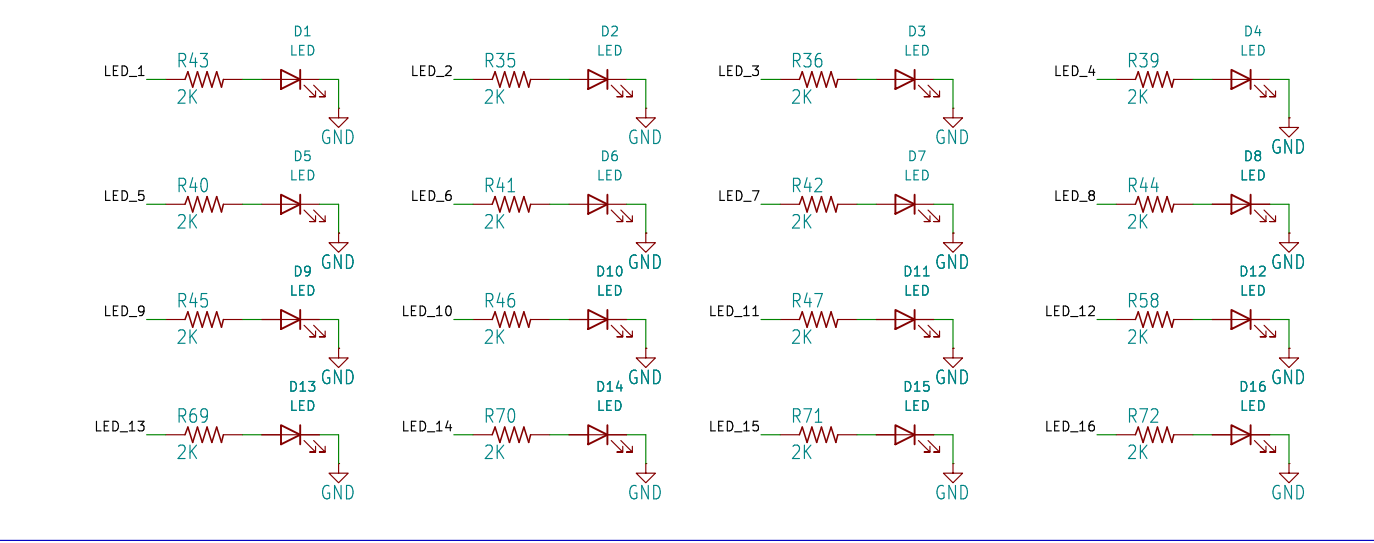


Not 100% sure why the PWR_FLAG is required here...

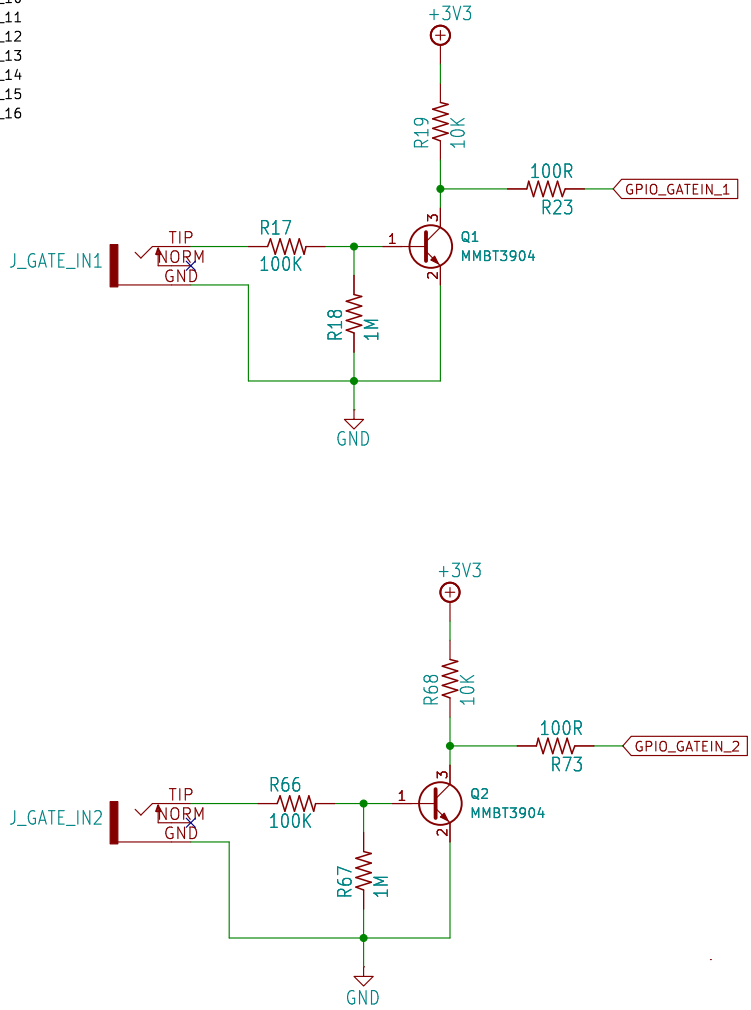
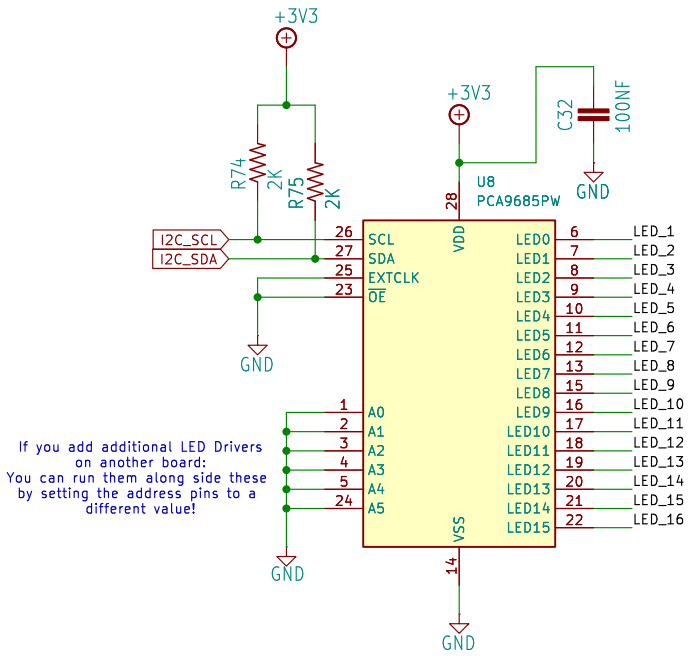
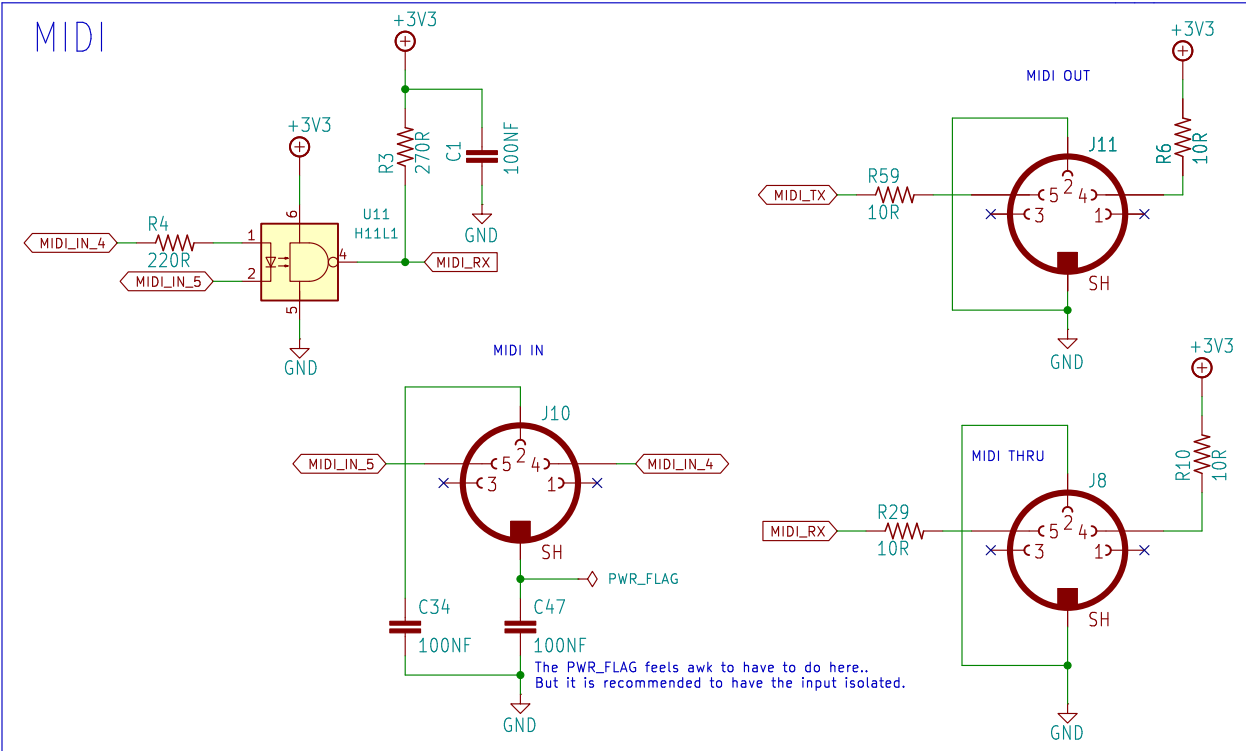
Switches + Pots



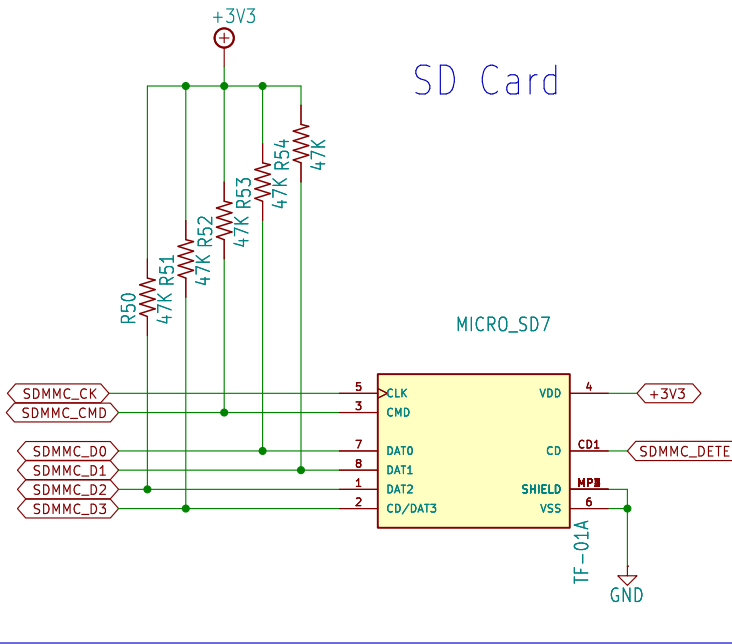
LEDs



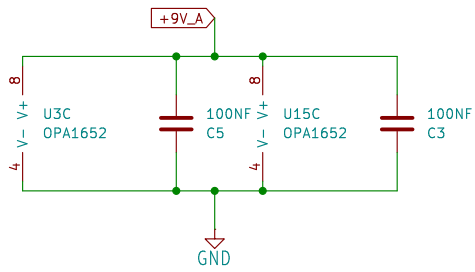
MIDI



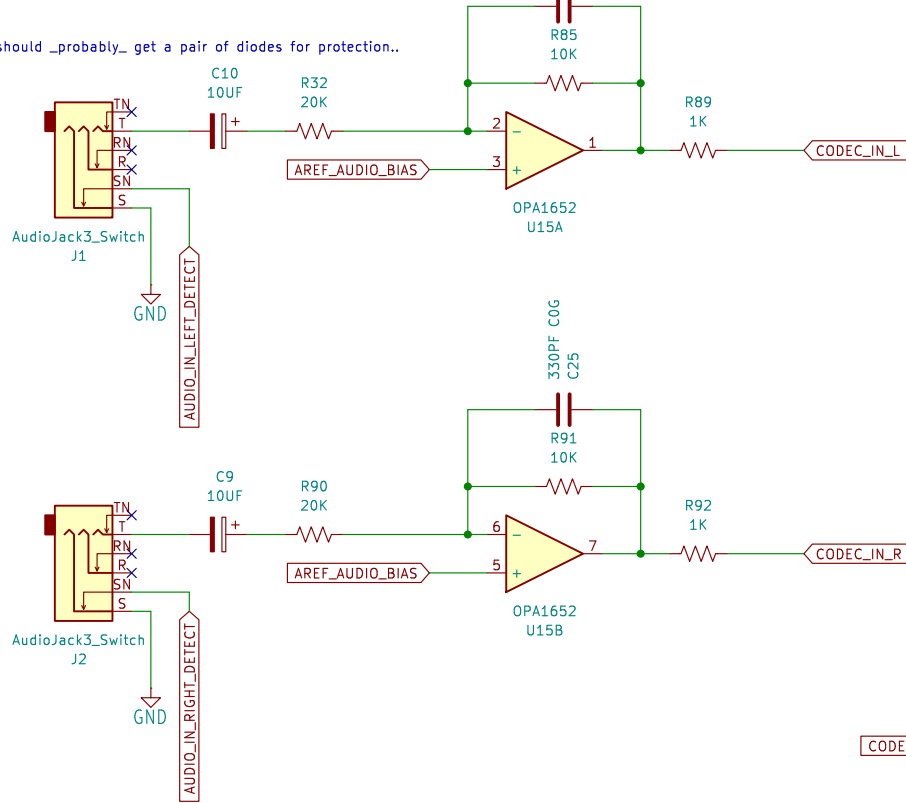
SD Card



AUDIO I/O



...s should _probably_ get a pair of diodes for protection..

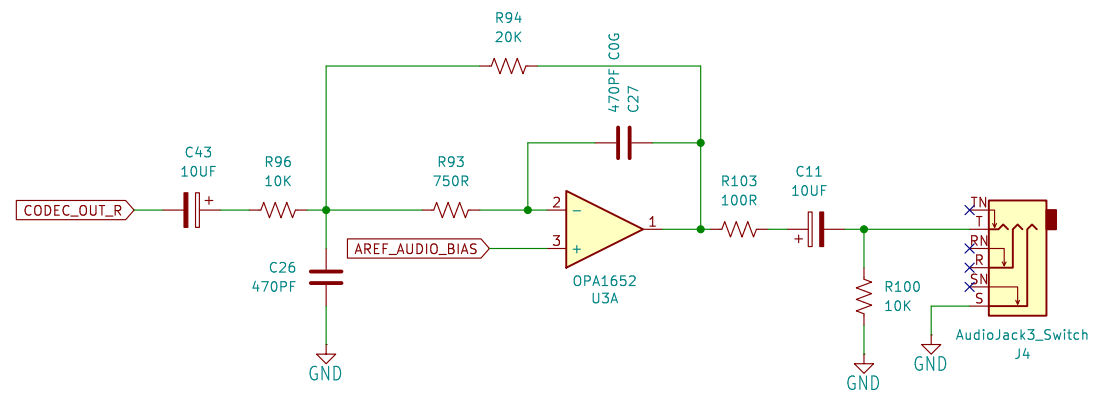


Gain is set to target +4dBu (maximum)
obviously this does not make it a pro-line-level device that way..
we'd need to probably have a certain amount of headroom.
If, instead we look at this as a -10dBV device, then we have
-14dB of headroom with these values.

$G_{in}=0.814$ ($R_{in}=10k$, $R_{fb}=16k$)
 $G_{out}=1.228$ ($R_{in}=10k$, $R_{fb}=24k$)

Alternatively we can go closer to the PCM3060's "2Vrms" reference and just do:
 $G_{in}=0.5$, $G_{out}=2$:shrug:

For now, I've implemented the left channel of audio with less atten/gain.

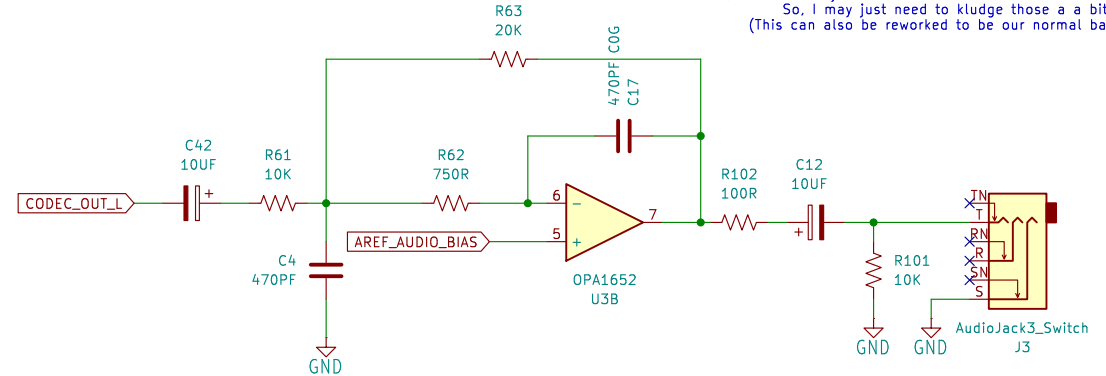


I updated these to be gain of 2.

However, I'm not really sure how the math works out for the other two values..

So, I may just need to kludge those a bit IRL.

(This can also be reworked to be our normal basic filter)



- DECISION – Two Choices:
1. Add LD1117 (or LP2985) or other 5V Linear Reg.
 2. Power TPA6110 from Seed's 3v3A

